Experiment No-10 : A* Algorithm

Aim

To write a Python program to implement the A* search algorithm in Artificial Intelligence.

Programming Code:

```
import heapq

def astar(graph, heuristic, start, goal):
    queue = [(heuristic[start], 0, start, [start])]
    best_cost = {start: 0}

    while queue:
        f, g, node, path = heapq.heappop(queue)
        if node == goal:
            return path, g

        for neighbour, cost in graph[node].items():
            new_g = g + cost
            if new_g < best_cost.get(neighbour, float("inf")):
                best_cost[neighbour] = new_g
                new_f = new_g + heuristic[neighbour]
                heapq.heappush(
                    queue, (new_f, new_g, neighbour,
                            path + [neighbour]))

    return None, float("inf")

graph = {
    'A': {'B': 1, 'C': 4},
    'B': {'A': 1, 'C': 2, 'D': 5},
    'C': {'A': 4, 'B': 2, 'D': 1},
    'D': {'B': 5, 'C': 1}
}
heuristic = {'A': 4, 'B': 2, 'C': 1, 'D': 0}

path, cost = astar(graph, heuristic, 'A', 'D')
print("A* Search")
print("Path:", " -> ".join(path) if path else "No path found")
print("Total cost:", cost)
```

OUTPUT:

Result

Thus, the A* algorithm was implemented successfully in Python. It uses path cost and heuristic estimates to find a least-cost path from the start node to the goal node.